6. Appendix: file formats

6.1 TPX3 raw file format

The .tpx3 raw data files contain the original data as sent by the readout board. It consists of chunks with an 8-byte chunk header prepended (see Table 6.1).
The chunk content consists of 8-byte words (little endian); the type of which is determined by the most significant nibble (i.e., the high nibble of the **last** byte):

- **TPX3 pixel data** with type 0xa for count_fb, or 0xb for the other modes, the maximum timestamp is 26.8435456 s;
- **SPIDR TDC data** with type 0x6, the maximum timestamp is 107.3741824 s;
- **TPX3 global time** with type 0x4, the maximum is ~81 days;
- **SPIDR control** with type 0x5.
- **TPX3 control** with type 0x7.

Sample code on how to decode this information from the raw data can be requested from your support contact.

Table 6.1. Chunk header

63 - 48 bit	47 - 40 bit	39 - 32 bit	31 – 0 bit			
Chunk size (bytes)	Reserved	Chip index	'3'	'X'	'P'	'T'
			"TPX3"			

Table 6.2. Pixel data packet (0xa, 0xb)

63 - 60 bit	59 - 44 bit	43 - 30 bit	29 - 20 bit	19 – 16 bit	15 – 0 bit
0xa (in count_fb)	PixAddr (see Table 6.6)	Integrated ToT (25ns)	EventCount	HitCount	SPIDR time (0.4096ms)
0xb (in other modes)	PixAddr (see Table 6.6)	ToA (25ns)	ToT (25ns)	FToA (-1.5625ns)	SPIDR time (0.4096ms)

Table 6.3. TDC data packet (0x6)

63 - 60 bit	59 – 56 bit	55 - 44 bit	43 - 9 bit	8 - 5 bit	4 – 0 bit
0x6	0xf = TDC1 Rise 0xa = TDC1 Fall 0xe = TDC2 Rise 0xb = TDC2 Fall	Trigger count	Timestamp (3.125ns)	Fine timestamp Takes value 1-12 (260.41666ps*)	Reserved

* Value 0 is an error state. Therefore: value 1 = 0ps, 2 = 260.41666ps, 3 = 520.83332ps, … 12 = 2.86458ns

Table 6.3. Global time data packet (0x4)

63 - 56 bit	55 – 48 bit	47– 16 bit	15 – 0 bit
0x44 = Time Low	Reserved	Timestamp (25ns)	SPIDR time (0.4096ms)

63 - 56 bit	55 – 32 bit	31 – 16 bit	15 – 0 bit
0x45 = Time High	Reserved	Timestamp (107.374182s)	SPIDR time (0.4096ms)

Table 6.4. SPIDR control packet (0x5)

63 - 56 bit	55 – 48 bit	47 – 0 bit
0x50 = Packet ID	Reserved	Packet count

64 – 60 bit	59 – 56 bit	55 – 46 bit	45 – 12 bit	11 – 0 bit
0x5	0xf = open shutter 0xa = close shutter 0xc = heartbeat	Reserved	Timestamp (25ns)	Reserved

Table 6.5. TPX3 control packet (0x7)

63 - 56 bit	55 – 48 bit	51 – 0 bit
0x71	0xa0 = End of sequential readout 0xb0 = End of data driven readout	Reserved

Table 6.6. Pixaddr format

15 - 9 bit	8 - 3 bit	2 – 0 bit
Double column There are 128 double columns on the chip, indexed left to right.	Super pixel There are 64 super pixels in each double column, stacked vertically, indexed bottom to top.	Pixel index There are 8 pixels in each super pixel. Indexed bottom to top, left to right. 0-3 = left column, 4-7 = right column.